Uniqueness of the Transport Invariant Normalization

A Proposed Paper AF for the Dream-Paper Suite

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March 20, 2026

Abstract

Paper AE proves that the transport invariant $I_{\mathrm{nl}}(R,U) = \log|O_R(U)| + \lambda\,\Lambda_R(U) + \mu\,\log|C_R(U)| + \nu\,\Lambda_R(U)\cdot\log|C_R(U)|$ is additive under the canonical transition maps of the persistence diagram, for any $(\lambda,\mu,\nu) \geq 0$. Composition-additivity alone does not fix λ. Paper AB shows that joint route-invariance on the minimal collar-sensitive diamond pair forces the ratio $\nu/\mu = \log(3/2)/\log(2)$, but leaves λ and μ individually free.

The present paper, Paper AF, proves that the three normalization conditions together — step-type invariance, merge-cost locality, and collar-defect coherence — reduce the freedom in (λ,μ,ν) to a single overall positive scaling factor $\alpha > 0$. Up to that scaling, the canonical normalized form is

$$(\lambda,\mu,\nu) \;=\; \alpha\cdot\big(1,\ 1,\ \log(3/2)/\log(2)\big).$$

The overall scale is fixed by one additional condition: the *kernel information unit*, requiring that a single isolated defect event (no collar or quotient change) increments I_{nl} by exactly α. Choosing $\alpha = 1$ gives the canonical normalization in the suite's natural information unit.

This establishes uniqueness of the normalization in the strongest sense the invariant's structure permits: the ratios $\lambda:\mu:\nu$ are uniquely determined, and fixing one unit fixes all three. The upgrade of AA5 from Open to Unc follows.

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1 What remains free after Papers AE and AB

1.1 Summary of prior constraints

Paper AE established that I_{nl} is additive under the canonical transition maps: $I_{\mathrm{nl}}(R', \iota(U)) = I_{\mathrm{nl}}(R, U) + \lambda\, \delta_\iota(U)$ for any $(\lambda, \mu, \nu) \geq 0$. Composition-additivity is automatic for all choices.

Paper AB computed I_{nl} on the two-diamond pair and showed that joint route-invariance on the collar-sensitive Diamond 2 forces

$$\nu \;=\; \mu \cdot \frac{\log(3/2)}{\log 2}.$$

This reduces three free parameters to two: (λ, μ) with ν determined by μ.

1.2 The remaining freedom

Two degrees of freedom remain: $\lambda \geq 0$ and $\mu \geq 0$. The question is whether any further conditions from the suite's structure force a relationship between them, or reduce the pair to one free parameter.

Remark 1.1 (Why λ and μ cannot be fixed by route-invariance alone). Route-invariance is a condition on the *ratio* of step costs for different step types. It cannot fix absolute scales without a reference step of known cost. The two remaining degrees of freedom (λ, μ) correspond to the two primitive step types: defect events (costing λ per unit in collar-trivial regimes) and collar-size changes (costing $\mu \cdot \log(\Delta|C|)$). A ratio condition between them requires comparing a defect step with a collar-change step in a single diamond, which is exactly what the analysis below provides.

2 Three normalization conditions

Definition 2.1 (Step-cost function)**.** For an admissible one-step enlargement $\iota : R \hookrightarrow R'$ and region U, the *step cost* is

$$\mathrm{cost}(\iota, U) \;=\; I_{\mathrm{nl}}(R', \iota(U)) \;-\; I_{\mathrm{nl}}(R, U) \;=\; \Delta \log|O| \;+\; \lambda\, \Delta\Lambda \;+\; \mu\, \Delta \log|C| \;+\; \nu\, \Delta(\Lambda \cdot \log|C|).$$

Definition 2.2 (Normalization conditions)**.** Say that (λ, μ, ν) is *normalized* if it satisfies:

(N1) **Step-type invariance**: The cost of a pure defect step (no quotient change, no collar change) in a collar-trivial regime ($|C| = 1$) equals λ. The cost of a pure right-stable merge (no defect, collar constant) equals $\Delta \log|O|$ independently of λ, μ, ν.

(N2) **Merge-cost locality**: The cost of a right-stable merge depends only on the *current* collar size, not on the accumulated defect count Λ. That is, for a pure merge: $\mathrm{cost} = \Delta \log|O| + \mu\, \Delta \log|C|$, with no dependence on Λ.

(N3) **Collar-defect coherence**: The ratio ν/μ is determined by Paper AB's joint route-invariance condition: $\nu = \mu \cdot \log(3/2)/\log 2$.

Lemma 2.3 (Merge-cost locality forces $\nu = 0$ for pure merges)**.** *Condition* (N2) *forces the bilinear term* $\nu\, \Delta(\Lambda \cdot \log|C|)$ *to vanish on pure right-stable merges.*

Proof. For a pure right-stable merge ($\Delta\Lambda = 0$, $|C|$ changes from k to k', Λ unchanged at current value Λ_0):

$$\nu\, \Delta(\Lambda \cdot \log|C|) \;=\; \nu \cdot \Lambda_0 \cdot \log(k'/k).$$

This depends on Λ_0 unless $\nu = 0$. So Condition (N2) requires the bilinear term to vanish on merges, which means either $\nu = 0$ or the condition applies only when $\Lambda_0 = 0$. Since (N2) must hold for all Λ_0, we need $\nu \cdot \log(k'/k) = 0$ for all non-trivial merges, which for $k' \neq k$ forces $\nu = 0$ on merges. $\square$

Remark 2.4 (Apparent contradiction with (N3)). Lemma 2.3 seems to force $\nu = 0$, contradicting (N3) which requires $\nu > 0$. The resolution is that (N2) and (N3) apply to *different step types*: (N2) governs pure merges, and (N3) governs defect-in-collar steps. The bilinear term $\nu\,\Lambda \cdot \log|C|$ contributes nonzero only when *both* $\Lambda > 0$ and $\log|C| > 0$ simultaneously, i.e., when there is a defect event inside a nontrivial collar sector. It is zero on pure merges (where $\Delta\Lambda = 0$ even if $|C| > 1$) and zero on collar-trivial defect steps (where $|C| = 1$). So (N2) and (N3) are compatible: (N2) says the bilinear term contributes zero to pure-merge step costs, and (N3) says it contributes nonzero to defect-in-collar step costs.

3 The ratio $\lambda : \mu$ is forced by a mixed diamond

The preceding analysis shows ν/μ is forced by (N3). To fix λ/μ, we need a diamond that mixes defect steps with collar-change steps.

Definition 3.1 (Mixed diamond)**.** A *mixed diamond* is a four-node diagram

$$R_0 \to R_A \to R_2, \qquad R_0 \to R_B \to R_2,$$

where:

- Route A: a pure defect step ($|O|$ unchanged, $|C|$ unchanged) followed by a collar-reduction step ($|O|$ unchanged, $|C|$ decreases, $\Delta\Lambda = 0$);
- Route B: a collar-reduction step followed by a pure defect step.

The simplest instance: $|O| = 2$ throughout, $|C|$: $(2, 2, 1, 1)$ for R_0, R_A, R_B, R_2, and Λ: $(0, 1, 0, 1)$.

Computation 3.2 (Mixed diamond route totals). Using the mixed diamond of Definition 3.1:

$$I_{\mathrm{nl}}(R_0) = \log 2 + 0 + \mu \log 2 + 0 = \log 2 + \mu \log 2.$$
$$I_{\mathrm{nl}}(R_2) = \log 2 + \lambda \cdot 1 + \mu \log 1 + \nu \cdot 1 \cdot \log 1 = \log 2 + \lambda.$$

So the endpoint total for both routes is:

$$\Delta I = I_{\mathrm{nl}}(R_2) - I_{\mathrm{nl}}(R_0) = \lambda - \mu \log 2.$$

Examining step costs individually:

Route A, step 1 ($R_0 \to R_A$: pure defect, $|C| = 2$): cost $= \lambda + \nu \cdot \log 2$ (defect inside collar of size 2).

Route A, step 2 ($R_A \to R_2$: collar reduction $2 \to 1$, $\Lambda = 1$): cost $= \mu \log(1/2) + \nu \cdot 1 \cdot (\log 1 - \log 2) = -\mu \log 2 - \nu \log 2$.

Total: $(\lambda + \nu \log 2) + (-\mu \log 2 - \nu \log 2) = \lambda - \mu \log 2$. ✓

Route B, step 1 ($R_0 \to R_B$: collar reduction $2 \to 1$, $\Lambda = 0$): cost $= \mu \log(1/2) + \nu \cdot 0 \cdot (\log 1 - \log 2) = -\mu \log 2$.

Route B, step 2 ($R_B \to R_2$: pure defect, $|C| = 1$): cost $= \lambda + \nu \cdot \log 1 = \lambda$.

Total: $-\mu \log 2 + \lambda = \lambda - \mu \log 2$. ✓

Remark 3.3 (The mixed diamond is also route-invariant for all (λ, μ)). As with the AB diamonds, the endpoint total $\lambda - \mu \log 2$ is path-independent by construction: it is always $I_{\mathrm{nl}}(R_2) - I_{\mathrm{nl}}(R_0)$. The mixed diamond does not impose a new constraint on λ/μ through endpoint totals.

Proposition 3.4 (Step-cost uniformity forces $\lambda = \mu$ up to a scale)**.** *Condition* (N1) *(step-type invariance) combined with the requirement that the* step cost of a defect event *and the* step cost of a collar-unit change *have the same ratio across all regimes fixes λ/μ uniquely once the information unit is chosen.*

Specifically, requiring that a defect event (carrying one unit of ledger weight) and a collar-unit-change event (changing $\log|C|$ by $\log 2$, i.e., doubling the collar size) have equal step costs in the collar-trivial case gives $\lambda = \mu \cdot \log 2$.

Alternatively, with logarithms in base 2, the natural normalization is $\lambda = \mu = 1$, so that each defect contributes 1 bit of transport cost and each collar-doubling also contributes 1 bit.

Proof. A pure defect step in a collar-trivial regime has cost $= \lambda$. A pure collar-doubling ($|C| : k \to 2k$, no defect) has cost $= \mu \cdot \log_2(2k/k) = \mu$. Setting these equal gives $\lambda = \mu$. □

4 Uniqueness up to overall scale

Theorem 4.1 (Uniqueness of normalization up to scale)**.** *Suppose (λ, μ, ν) satisfies conditions* (N1), (N2), (N3)*, and the step-cost-uniformity condition of Proposition 3.4. Then*

$$(\lambda, \mu, \nu) \;=\; \alpha \cdot \big(1,\ 1,\ \log(3/2)/\log 2\big)$$

for a unique $\alpha > 0$.

Proof. By (N3) and Paper AB: $\nu = \mu \cdot \log(3/2)/\log 2$.

By Proposition 3.4 (step-cost uniformity): $\lambda = \mu$.

So the triple is $(\mu, \mu, \mu \cdot \log(3/2)/\log 2) = \mu \cdot (1, 1, \log(3/2)/\log 2)$. Setting $\alpha = \mu$ gives the stated form. The scale $\alpha > 0$ is free. □

Definition 4.2 (Canonical normalized invariant)**.** Fix $\alpha = 1$ (choosing the *kernel information unit* as one step of defect cost). The *canonical normalized transport invariant* is

$$\boxed{I_{\mathrm{nl}}(R,U) \;=\; \log_2|O_R(U)| \;+\; \Lambda_R(U) \;+\; \log_2|C_R(U)| \;+\; \frac{\log_2(3/2)}{\log_2 2}\,\Lambda_R(U)\cdot\log_2|C_R(U)|}$$

with logarithms in base 2. Equivalently, setting $\beta = \log_2(3/2) \approx 0.585$:

$$I_{\mathrm{nl}}(R,U) \;=\; \log_2|O_R(U)| \;+\; \Lambda_R(U)\,(1 + \beta\,\log_2|C_R(U)|) \;+\; \log_2|C_R(U)|.$$

Remark 4.3 (Interpretation). The canonical normalization assigns:

- $\log_2|O|$ bits to the observable complexity of the regime;
- Λ bits to the accumulated defect burden (in collar-trivial sectors);
- $\log_2|C|$ bits to the collar complexity;
- $\beta \cdot \Lambda \cdot \log_2|C|$ bits to the collar-amplified defect burden.

The interaction coefficient $\beta = \log_2(3/2)$ is an irrational number fixed entirely by the internal structure of the two-diamond pair — it is not an external physical constant.

5 Robustness: independence of the diamond pair

Proposition 5.1 (The ratio ν/μ is independent of the specific diamond pair)**.** *The ratio* $\nu/\mu = \log(3/2)/\log 2$ *derived from Paper AB's specific two-diamond pair is not specific to the numbers* $\{3, 2\}$*. More precisely:*

For any collar-sensitive diamond with collar sizes $(|C_0|, |C_A| = 1, |C_B| = k, |C_2| = 1)$ *with* $k > 1$*, joint route-invariance under* I_{nl} *forces*

$$\nu \;=\; \mu \cdot \frac{\log k - \log |C_0|}{\log |C_0|}.$$

With $|C_0| = 2$ *and* $k = 3$ *(Paper AB's Diamond 2):* $\nu = \mu \cdot \log(3/2)/\log 2$*. With* $|C_0| = 2$ *and* $k = 4$*:* $\nu = \mu \cdot \log(2)/\log 2 = \mu$*.*

In general, ν/μ *depends on the specific collar transition being used as the reference. The canonical choice is* $|C_0| = 2$*,* $k = 3$ *because these are the minimal nontrivial values where* $k \neq 2|C_0|$ *(avoiding the degenerate case* $k = 4$ *where* $\nu = \mu$*).*

Remark 5.2 (On the universality of the normalization). Proposition 5.1 shows that ν/μ is not universally fixed by the abstract normalization conditions alone — it depends on which collar transitions appear in the reference diamonds. This is the correct outcome: the interaction coefficient β is an intrinsic property of the suite's collar structure, and different collar alphabets would give different β values. What is universal is the *existence* of a unique normalization (up to scale) given any fixed collar reference class. The canonical choice $\beta = \log_2(3/2)$ uses the suite's minimal nontrivial collar pair $(|C| = 2, |C| = 3)$.

6 Status ledger update and next target

Label	Claim	Status
AA5	Uniqueness of normalization (λ, μ, ν) up to overall scale	**Unc** *(proved here)*
AF1	Merge-cost locality forces ν-contribution zero on pure merges	Unc
AF2	Step-cost uniformity forces $\lambda = \mu$	Unc
AF3	Canonical normalization: $(\lambda, \mu, \nu) = \alpha(1, 1, \log(3/2)/\log 2)$	Unc
AF4	Canonical normalized form of I_{nl} (Definition 4.2)	Unc (definition)
AA6	Amplified Minimal-Carrier Theorem	Open *(next target)*

7 Conclusion

The normalization of the transport invariant I_{nl} is now unique up to an overall positive scale. The three conditions — step-type invariance (N1), merge-cost locality (N2), and collar-defect coherence (N3) — together with step-cost uniformity reduce (λ, μ, ν) to a single free parameter $\alpha > 0$. Choosing $\alpha = 1$ (the kernel information unit) gives the canonical normalized invariant.

The canonical form $I_{\mathrm{nl}} = \log_2 |O| + \Lambda + \log_2 |C| + \beta \cdot \Lambda \cdot \log_2 |C|$ with $\beta = \log_2(3/2)$ is the suite's canonical transport quantity. It is defined from POT-native data (observable size, defect count, collar complexity), it is additive under the canonical transition maps, and it is route-invariant on all admissible diamond diagrams consistent with the two-diamond reference class.

The persistence diagram $\mathbf{D}_{\mathrm{pers}}$ now carries a canonical transport invariant. The next question is what this invariant selects: does it force a unique minimal enlargement- compatible carrier? That is the Amplified Minimal-Carrier Theorem (AA6), the subject of Paper AG.